Client Onboarding Agent: Design Document

Summary

The fund onboards about 15 new clients a month with one employee, and wants to reach 30 without hiring. Two things hold onboarding back: clients drop out when the portal blocks them for a missing document, and the employee fills a 30-question form for the banks by hand from each client's documents.
We propose an onboarding agent that takes over the repetitive work while the employee keeps every decision:

- Clients can submit what they have; the agent welcomes them, tracks what is missing and follows up by email until the file is complete.
- The agent reads the bank statement, tax return and portfolio statement, and drafts the bank form. Every answer links to the sentence it came from, and anything not in the documents is marked as missing rather than guessed.
- The employee reviews, corrects and approves the form; only then is it sent to the bank. Bank replies are read and turned into next steps.
- A single status board shows every client from "started" to "onboarded".

The agent runs on Prem's confidential inference, so client documents are processed in a protected environment in the EU. We start in shadow mode next to the current process and switch to assisted mode once the drafted forms match the employee's on a test set of past onboardings.

What we learned in discovery

The bottleneck is one person doing manual form work, plus clients abandoning incomplete submissions.

Topic	**What the client told us**
Volume	About 15 clients onboarded per month; target 30. Demand peaks in the first week of the month.
People	One employee runs onboarding and is not familiar with AI tools. 4–5 people check manually that all steps were followed.
Current workflow	Work with the client, collect documents, fill the bank form, send it to the bank, wait for the bank's answer, sign, client onboarded.
Documents	Bank statement, tax statement, investment portfolio statement.

Bank form	About 30 questions, the same for every bank (e.g. total spent on physical assets). The employee fills each answer from the documents; the documents may not contain every answer.
Main pain point	The portal won't let clients submit without every document; clients with short attention spans drop out.
Tools	Gmail, Excel, Google Drive for documents. The company uses AWS.
Data	Personal and financial data; some must stay in the EU, some in Italy. The agent should have the same access as the employee.
Hardware	Some RTX 6000 GPUs that are not used today.
Out of scope	Asset classes and allocation: they change and are not part of onboarding.
Availability	Onboarding isn't continuous; downtime can be coordinated outside peak periods.

Goals and success metrics

The goal is throughput without losing quality: double the clients onboarded per month with the same team, and no increase in forms the banks send back.

Metric	**Why it matters**	**Target (to confirm in the pilot)**
Clients onboarded per month	The business goal	From 15 to 30
Employee time per client	What limits throughput today	Roughly halved; baseline measured in shadow mode
Drop-off rate	Clients lost before the file is complete	Lower than today; baseline to measure
Time from first upload to bank-ready form	Speed the client feels	Same day once documents are complete
Form field accuracy	Wrong answers reach the bank	At least as good as the employee on the test set
"Check needed" replies from banks	Proxy for form quality	No increase versus today

Steps followed (reviewers' checklist)	What the 4–5 reviewers check	100%, visible in the audit log

We agree the exact targets with management after measuring the current baseline, not before.

Scope of the first phase

The first phase automates preparation and follow-up; every decision and everything sent to a bank stays with the employee.

In scope:

- Client upload portal that accepts partial submissions and saves progress.
- Welcome email and automatic, templated reminders for missing documents.
- Reading the three document types and drafting the 30-question bank form, with the source of every answer.
- Targeted questions to the client for answers the documents don't contain.
- Employee review and approval of the form; export to Excel in the bank's format.
- Reading bank replies and proposing the next step.
- Status board and audit log for the employee and the reviewers.

Out of scope for now:

- Asset classes and investment allocation (dynamic, decided elsewhere).
- Sending anything to a bank without the employee's approval.
- Contract signing (stays in the current process).
- Identity verification and anti-money-laundering decisions: the agent can collect documents for them, but the checks remain human.
- Running models on the in-house GPUs (a later option, see the rollout plan).

Proposed workflow and statuses

The workflow follows the employee's current steps; the agent does the reading, drafting and chasing, and the employee approves at the two points that matter: the form and anything a bank asks for.

Status	**What happens**	**Who acts**	**Next**
Started	Client submits the first documents; welcome email sent	Client, agent	Awaiting documents or Documents complete
Awaiting documents	Agent lists what is missing and sends reminders (e.g. after 1 and 3 days)	Agent, client	Documents complete
Documents complete	Bank statement, tax statement and portfolio statement all received	Agent	Form drafted

Form drafted	Agent fills the 30 answers with sources; answers not in the documents become questions to the client	Agent, client	In review
In review	Employee checks each answer, corrects it, approves the form	Employee	Sent to bank
Sent to bank	Approved form exported in the bank's format and sent	Employee	Bank accepted or Bank needs information
Bank needs information	Agent reads the reply, lists what the bank asks for, drafts the request to the client	Agent, employee	In review
Bank accepted	Employee confirms; signing follows the current process	Employee	Onboarded

The key change from today: a missing document no longer blocks the client. They submit what they have, and the agent keeps the file moving.

Architecture

A fixed workflow runs the steps; models work inside individual steps, and the only path to a bank goes through the employee.
Flow: client upload portal → Google Drive folder per client → onboarding workflow (document reader, form filler and verifier, email drafter) → employee review → bank. Bank replies come back into the workflow, and every step is written to the status board and audit log. All model calls go to Prem's confidential inference.

Component	**How it works**
Upload portal	Simple web form; files saved to a Google Drive folder per client; partial submissions allowed
Onboarding workflow	Deterministic steps with one status per client and a job queue; model calls happen only inside steps
Document reader	Text extraction for digital PDFs, a vision

	model for scans and photos; identifies document type and year
Form filler	A language model fills the 30 answers into a fixed schema: value, source document, quoted text, or "not found"
Verifier	Rules, not a model: each quote must appear in the source document; conflicting values across documents are flagged
Email drafter	Fixed templates for welcome and reminders; the model drafts targeted questions about missing answers
Bank reply reader	The model classifies a reply as accepted or information requested and lists what the bank wants
Tools	MCP servers for Google Drive, Gmail and Excel or Sheets, using a service account scoped to onboarding
Models	Open-weight models on Prem's confidential inference: start with a large model to set the quality bar, then test a smaller, possibly fine-tuned model
Status board and audit log	A small database the employee and reviewers read; every step, source, edit and approval is recorded

Why a workflow rather than a free-running agent. Onboarding follows the same steps for every client, and the reviewers check that every step was followed. A fixed workflow is predictable, easy to audit and cheaper. The model's judgement is used where inputs vary: reading documents, filling answers and drafting messages.

Human in the loop and edge cases

The employee isn't familiar with AI tools, so there is no chat to learn: the agent presents work as a checklist to approve.

What the employee sees:

- A status board with every client, their status and what is waiting on whom.
- A review screen for each form: every answer shows its value, the document and sentence it came from, and a flag if it couldn't be verified. The employee edits or approves each answer, then approves the form.

What the agent may do on its own, and what needs approval:

Action	**Phase 1**

Welcome email and document reminders (fixed templates)	**Automatic**
Questions to the client about missing answers	**Drafted by the agent, sent after employee approval**
Sending the form to a bank	**Employee only**
Replying to a bank	**Employee only, from the agent's draft**
Changing a status to Onboarded	**Employee only**

Edge cases and how they are handled:

Case	Handling
A document never arrives	**Reminders on a schedule; after the last one, the employee is alerted to call the client**
An answer isn't in the documents	**Marked "not found" and turned into a question to the client; never estimated**
Two documents disagree (e.g. income on the tax return vs. the bank statement)	**Both values shown to the employee, who decides**
Wrong or outdated document (e.g. last year's tax return)	**Detected from the document type and date; the client is asked for the right one**
Poor scan or photo	**Read with a vision model; if still unreadable, the client is asked for a better copy**
The client replies by email with attachments	**The agent adds them to the client's file and re-checks completeness**
A document contains text that looks like instructions	**Treated as data only; the agent has no way to send to banks or change recipients**
Ambiguous bank reply	**Passed to the employee with the agent's reading of it**
The agent is unavailable	**The employee continues manually; the status board stays readable**

Evaluation and validation

Past onboardings are a ready-made test set: each one has the client's documents and the form the employee actually sent to the bank.

Test set. Collect the documents and final forms from past onboardings (pseudonymised), as many as are kept, covering different banks, document layouts and clients with missing

information. The employee confirms the correct answer for each field, including fields where the right answer is "not in the documents".

Metrics per step:

Step	Metric
Document recognition	**Share of documents correctly identified by type and year**
Completeness check	**Missing documents detected (a miss means a stuck file)**
Form answers	**Share of the 30 answers correct, per field; amounts within an agreed tolerance**
Not-found answers	**Share of "not found" that are truly absent; the agent must never invent a value**
Evidence	**Share of answers whose quoted source text really appears in the document (checked automatically)**
Bank replies	**Correct reading of accepted vs. information requested**
End to end	**Time to bank-ready form; employee minutes per client; steps followed**

Only a small model-based judge is needed: most answers are values that can be compared directly with the employee's.

Validation in production, in stages:

1. **Shadow mode:** the agent drafts forms in parallel while the employee works as today; we compare field by field and measure the employee's current time per client.
2. **Assisted mode:** the employee starts from the agent's draft and approves it; the share of answers the employee changes is the main quality signal.
3. **Steady state:** the reviewers keep their checklist, now read from the audit log; bank "check needed" replies are tracked against the pre-pilot rate.

Every change to prompts, models or rules is re-run on the test set before release, and is blocked if any metric drops.

SLA and scaling

The load is small and bursty, so the design favours reliability and simplicity over raw capacity.

Dimension	Commitment
Speed	Documents checked within minutes of upload; a drafted form within 15 minutes of the file becoming complete; reminders on a fixed schedule. Processing runs day and

	night, so uploads made in the evening are ready the next morning.
Availability	99.5% during business hours. Maintenance is planned outside the first-week peak.
Fallback	If the agent is down, the employee works as today from the documents in Google Drive. Nothing is lost; files resume where they stopped.
Load	30 clients a month is about 1–2 a day, with a peak in the first week of the month. With 3 documents and a few model calls per document, that's a few dozen calls a day.
Quality	The metrics in the evaluation section, reviewed monthly with management.

Scaling. At this volume, dedicated GPUs or GPU autoscaling aren't needed: a managed confidential inference API and a simple job queue cover the first-week peak, and cost is paid per use. If volumes grow or management wants everything in-house, a smaller model, fine-tuned on the confirmed forms, could later run on the company's existing RTX 6000 GPUs. That step is only worth it once the test set shows the smaller model is as accurate.

Security and compliance

Client documents are personal and financial data, and uploaded files are untrusted input, so protection covers both where data is processed and what the agent is allowed to do.

- **Confidential processing.** Model calls go to Prem's confidential inference in the EU. Documents are encrypted on the way in, decrypted only inside a Trusted Execution Environment, held only in memory, and not stored after processing. Hardware attestation lets the company verify what is running instead of relying on our word. The exact region and whether it can run inside the company's own AWS account are to be confirmed with Prem.
- **What this doesn't replace.** The company still needs a data protection impact assessment, retention rules and access control under GDPR; confidential computing protects data during processing only.
- **Data residency.** Documents stay in Google Drive as today; data that must stay in Italy or the EU is processed only in the matching region.
- **Least privilege.** The agent uses its own service account limited to the onboarding folder and an onboarding mailbox or label, not the employee's whole inbox and drive. It can send email only from fixed templates or after approval, and only to the client's registered address.
- **Untrusted documents.** Text extracted from uploads is treated as data, never as instructions; answers are validated against a fixed schema; the agent has no tool that could send to a bank or change a recipient.

- **Audit log.** Every step is recorded: document received, answer extracted with its source, edits and approvals with who made them, emails sent, model and prompt version. The 4–5 reviewers check steps from the log.
- **Regulation.** Onboarding in wealth management carries anti-money-laundering due diligence and investor suitability duties (MiFID II). The agent collects and organises the information; the assessments and the decisions remain with the company's staff.

Risks and mitigations

The biggest risk is a wrong answer reaching a bank; every mitigation below keeps a person between the agent and the bank.

Risk	**Mitigation**
The agent invents an answer that isn't in the documents	Every answer must quote its source; the quote is checked against the document automatically; anything unverified is flagged; the employee approves every form
Few past onboardings to test on	Start shadow mode early, so every new onboarding adds to the test set
The employee doesn't trust or adopt the tool	No chat to learn; a checklist with sources; shadow mode first, so they see it work before relying on it
Documents vary across banks and countries (layouts, languages)	Vision model for scans; test set covering each document source; unknown layouts sent to the employee
Clients ignore reminders	Short, friendly templated emails with a direct upload link; the employee is alerted after the last reminder
A bank changes its form	The form is configuration, not code; a change is a new version, re-run on the test set
Data protection or regulatory objections	Confidential processing in the EU, least-privilege access, audit log, and the impact assessment done at the start

Rollout plan and the proof of concept

Autonomy is earned step by step: each phase starts only when the previous one's numbers hold.

Phase	**What changes**	**Moves on when**
0. Proof of concept	The full flow on sample	Management and the

	clients, to agree on the design	employee agree the flow fits their work
1. Shadow mode	Agent drafts in parallel on real onboardings; the employee works as today	Drafted forms match the employee's on the test set, and the baseline is measured
2. Assisted mode	The employee starts from the agent's draft; welcome and reminder emails go live	Fewer employee minutes per client, no rise in bank "check needed" replies
3. Extensions	More automatic client follow-ups; direct submission to bank portals where they exist; optionally a smaller model on the in-house GPUs	Each extension passes the test set first